# SATYAPAN (VERIFICATION) — GRAND FINALE TECHNICAL MASTERCLASS

**Track 03: Deepfake Detection for Financial Communications · IIT Delhi Grand Finale (24–25 Oct 2026)**

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**EXECUTIVE THESIS & THE ARCHITECTURAL PIVOT:**

*"When the voice or video is a deepfake, the payment request is still real. Anchor on the payment floor; detectors can only raise the band."*

Indian citizens and SME finance teams lost over ■**22,845 Crore** to cyber and payment fraud in 2024 (Lok Sabha official reply, July 2025). Attacking vectors have converged: scammers combine digital arrest intimidation (fake CBI/police video calls) with voice clones and forged invoices. Classical standalone deepfake detectors fail catastrophically in production because: (1) VoIP and cellular codecs (Opus 16k, AMR-NB, GSM) destroy high-frequency acoustic artifacts (causing up to 50% AUROC loss), (2) injecting mere 10 dB background noise collapses voice clone recall from 63.9% down to 1.9%, and (3) in daily accounts payable, a 5% false-positive rate paralyzes legitimate commerce.

**The Satyapan Solution:** A deterministic policy engine compares every transaction against the company's vendor master or citizen baseline to set a trusted **floor** (PASS, VERIFY, or HOLD). Multi-modal detectors (speech asks, face blending CNNs, voice clone scorers, and on-device OCR) act as evidence that **can strictly raise the security band, never lower it**. No raw probability scores are ever exposed to the user, eliminating adversarial oracle-abuse tuning.

## 1. System Architecture & The 6 Non-Negotiable Invariants

Satyapan bridges two operational environments: **SME Finance Teams** (maker/checker payment approvals, vendor master reconciliation, and out-of-band mobile verification) and **Individual Citizens** (one-click scam scanning on /check and an ambient desktop companion **Satya** that monitors video calls only after explicit consent). All code strictly adheres to 6 mathematical invariants enforced by automated tests:

| Invariant Principle | What it Guarantees in Simple Words | Enforcement Mechanism & Test Suite |
|---|---|---|
| **1. Raise-Only Monotonicity** | Detectors can only escalate a security band (PASS → VERIFY → HOLD). No detector output, email, or LLM output can ever clear a payment or lower a band. Furthermore, heuristic/ML signals are hard-capped at VERIFY (they cannot freeze money alone without an ask or policy violation). | `Enforced by policy engine reducer.` `tests/invariants/test_monotonicity.py` `tests/invariants/test_llm_cap.py` |
| **2. Trusted Floor Isolation** | The baseline floor is computed strictly from trusted accounting records: maker-entered transaction details + vendor master + policy config. Untrusted external data (email body, invoice PDF text, call audio) is mathematically forbidden from entering compute_floor. | `Strict contract type segregation.` `tests/invariants/test_untrusted_floor.py` |
| **3. Zero-Mic & Consent First** | Satya starts asleep. Audio is monitored **only after explicit user consent** for that specific call. Satya captures **system audio loopback only** (what the laptop speaker plays—never the user's microphone). All transient audio buffers are wiped from RAM immediately upon call stop. | `Electron loopback capture harness.` `desktop/main/capture.ts` `tests/tasks/test_t25_companion.py` |
| **4. Local-First Privacy Boundary** | All processing (ASR, visual inference, OCR, policy evaluation) executes 100% locally on the device. The desktop app connects strictly to 127.0.0.1:8000. Network calls outside src/satyapan/adapters/ are completely blocked. | `Socket binding & import boundary checks.` `tests/invariants/test_network_boundary.py` `test_no_network.py` |
| **5. Oracle Coarsening** | End-users, makers, and attackers never see raw confidence scores, probabilities, or feature weights. They see only the coarse band and plain-text rationales. This prevents attackers from executing black-box gradient descent to optimize deepfakes against our models. | `API schema serializers coarsen floats.` `tests/invariants/test_oracle_coarsening.py` |
| **6. Human Review & No Auto-Report** | No code may automatically submit reports to national cybercrime portals (1930, NCRP), banks, or police. Reports are pre-filled drafts (submission = 'MANUAL_ONLY') cryptographically signed under Section 63 of the Bharatiya Sakshya Adhiniyam (BSA). | `Report serializers enforce MANUAL_ONLY.` `tests/invariants/test_no_auto_reporting.py` |

## 2. End-to-End Operational Workflows Explained

To understand how data flows through Satyapan in real time, examine the 3 core operational workflows:

| Operational Journey | Step-by-Step Data Flow & Engineering Pipeline | User Experience & Outcome |
|---|---|---|
| **Workflow 1: Live Call Guardian** (Citizen / Finance) | **1. Meeting Watcher:** Detects active call window (Meet, Zoom, WhatsApp, Teams).<br>**2. Consent Card:** Mascot wakes, requests explicit yes before listening.<br>**3. System Audio Capture:** PCM worklet grabs system loopback (no mic) at 16kHz, slices at pauses via local Silero VAD.<br>**4. Local Whisper ASR:** Generates captions ~1s behind speech on-device.<br>**5. Real-Time Asks:** Text stream evaluated every 2s for A1–A7 fraud asks.<br>**6. Video Stream:** Downsampled 15 fps video crops sent to GenD face CNN.<br>**7. Policy Action:** If threat detected, companion displays alert card. | Satya displays alert quoting exact words (*'CBI never asks for UPI'*). Offers 3 immediate buttons: **Copy Safe Reply & Hang Up**, **Call 1930**, or **Prepare Golden-Hour Report**. |

| | | |
|---|---|---|
| **Workflow 2:**<br>**Citizen Quick Check**<br>(Web / Mobile /check) | **1. Drag & Drop:** User drops suspicious SMS, WhatsApp forward, screenshot, photo, voice note, or video.<br>**2. Smart Dispatcher:** Detects MIME type; routes images to native OCR + face crop, audio to Whisper + XLS-R, text to de-obfuscation.<br>**3. Text Model Scoring:** Normalized tokens scored by TF-IDF scam model (12ms) + impossible claim rules (digital arrest, electricity cutoff, courier customs).<br>**4. Unified Verdict:** Engine outputs coarse verdict with plain English/Hindi rationale.<br>**5. Recovery Engine:** If money was already sent, prompts for transaction UTR. | Instant verdict in seconds. If user clicks *'Money already sent?'*, generates an instant 1930 call script with the UTR on line 1, a formal bank freeze letter, and NCRP portal fields. |
| **Workflow 3:**<br>**SME Payment Verification**<br>(Maker / Checker / OOB) | **1. Payment Entry:** Maker enters invoice payment details (amount, IFSC, account).<br>**2. Deterministic Floor:** Engine checks vendor master. Existing beneficiary & routine amount → PASS. Changed bank account → VERIFY. Unknown account → HOLD.<br>**3. Multi-Modal Corroboration:** Inbound PDF invoice parsed by PDFPlumber; Screen Guard checks accounting portal.<br>**4. X1 Cross-Channel Signal:** Correlates 30-min call urgency with wire request.<br>**5. OOB Confirmation:** QR code displayed on screen routes to approver's phone over local LAN. Approver taps *'Yes, approved'* or *'Not me'*. | If approver denies: Transaction immediately frozen at HOLD. Full Section 63 BSA legal evidence pack compiled with SHA-256 chain of custody. |

## 3. Multimodal Node Deep-Dive: Exactly How Each Node Works

Every node in Satyapan's multimodal architecture is engineered to be modular, transparent, and resilient to real-world edge cases. Here is the exact technical operation of every individual node:

| Node Name | Input & Preprocessing | Mathematical & Algorithmic Transformation | Output Signal & Platform Support |
|---|---|---|---|
| **Node 1:**<br>**Screen Guard OCR**<br>(Cross-Platform) | Full-screen grab taken during active sessions. Satya's own windows are excluded via OS window ID exclusion so it never reads its own alerts. | **macOS:** Native Apple Vision VNRecognizeTextRequest (.fast level) in Swift helper binary.<br>**Windows:** In-memory PowerShell WinRT projection calling Windows.Media.Ocr.OcrEngine via .NET runtime (zero third-party tools!).<br>**Linux:** Desktop Electron capture piped to CLI tesseract. | Extracts plain text, amounts, and IFSC bank codes.<br>Triggers SIGNAL_SCREEN_TEXT. |
| **Node 2:**<br>**Text De-Obfuscator & Scam Model** | Raw text from OCR, emails, pasted messages, or Whisper speech transcripts. | **Layer 1 (De-obfuscation):** NFKD Unicode normalization, Cyrillic homoglyph replacement (e.g. Cyrillic '■' → Latin 'a'), leet-speak mapping (0→o, 1→l, @→a), stripping zero-width joiners/spaces, collapsing whitespace.<br>**Layer 2 (Classifier):** TF-IDF n-grams (1-2 words) + Logistic Regression (C=1.0, 5-fold CV). Calibrated on pooled validation sets for 1% FPR. | TEXT_SCAM_LIKELY (ML signal, capped at VERIFY).<br>Latency: **12 ms** on CPU. |
| **Node 3:**<br>**A1–A7 Contextual Ask Extractor** | De-obfuscated text from messages or live Whisper transcription chunks. | Deterministic context-aware parser operating across English, Hindi, and Hinglish. Uses phrasebook patterns coupled with context filters to reject false positives (e.g., distinguishing 'Your OTP is 123456, do not share' from 'Please share the OTP with our agent'). Identifies digital arrest, safe custody accounts, remote tool installations, and secret wires. | Emits ASK_OTP, ASK_NEW_BENEFICIARY, ASK_SECRECY, ASK_REMOTE_ACCESS.<br>Raises band to VERIFY or HOLD. |
| **Node 4:**<br>**YuNet Tracker & Face Extractor** | Downsampled 15 fps video frames from screen capture window or uploaded MP4/WebM. | Stock face detectors (Haar, standard SSD) fail on 53.5% of compressed call video faces. We deploy a dual-stage pipeline: (1) YuNet lightweight CNN localizes small faces down to 20x20 pixels at 360p, (2) MediaPipe facial mesh extracts 468 landmark points to align face crops to canonical orientation. Tracks facial bounding boxes across consecutive frames. | Canonical 224x224 aligned face crops + facial landmark vectors.<br>Handles small faces at 360p. |
| **Node 5:**<br>**GenD Video CNN & Lip-Sync Scorer** | Aligned 224x224 face crops + synchronized audio track. | **GenD CNN (MIT):** Evaluates boundary blending artifacts and frequency anomalies across 6-second sliding temporal windows (3s hop).<br>**Consecutive-Window Rule:** Requires 2 consecutive windows over threshold to trigger alert, suppressing single-frame lighting artifacts.<br>**Lip-Sync:** Measures Pearson correlation between mouth open/close aperture delta and audio speech energy envelope. | VIDEO_LIKELY_SYNTHETIC (+1 band) or VIDEO_INCONCLUSIVE.<br>Latency: **1.3s**/crop. |
| **Node 6:**<br>**Voice Clone XLS-R Scorer** | Raw audio stream (WAV, Opus, or GSM container). | Audio partitioned into 3.0s windows. Sliced into 16kHz mono PCM. Evaluated by self-supervised Wav2Vec2-Large-XLSR ONNX trunk. Extracts 1024-dimensional acoustic embeddings, projected through a trained single linear classifier head. Gate enforces >= 3s speech (audio < 3s abstains with 'too short to judge' rather than guessing). | VOICE_SUSPICIOUS (VERIFY) or VOICE_CLONE_LIKELY.<br>Latency: **0.4s**/window. |
| **Node 7:**<br>**X1 Cross-Channel Correlator** | Time-series signal ledger across audio, screen OCR, and accounting events. | Evaluates temporal correlation within a 30-minute rolling sliding window. Correlates external urgency signals (A4 urgent threats, A5 remote access, or call asks) with concurrent financial actions (maker wire transfer, new beneficiary entry). If an ask co-occurs with an accounting change, the combined X1 signal escalates directly to HOLD, even if individual detectors were at VERIFY. | Emits composite signal X1_CROSS_CHANNEL_ATTACK.<br>Deterministic rule execution. |

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|---|---|---|---|
| **Node 8:**<br>**Living Mascot & Copilot Shell** | Engine SSE event bus streaming decisions, signals, and transcript updates. | Procedural Three.js mascot **Satya** with 12 distinct emotional poses (sleeping, watching, alert, gulp, thinking) and spring-physics gaze tracking. Delivers trilingual bubble cards (English, Hindi, Hinglish). Manages OOB QR confirmation generation and auto-compiles the golden-hour report pack. | Electron 4-window manager.<br>86 Vitest tests green.<br>60 fps smooth animation. |

## 4. The Model Tournament: What We Trained, Compared, and Shipped

**We did not blindly download a model from Hugging Face and claim it worked.** Every model in Satyapan underwent a rigorous pre-registered tournament against our own trained architectures and off-the-shelf baselines under real call compression:

| Domain | Model / Architecture Evaluated | Role & Training Provenance | Benchmark Metric (Val/Test) | Selection Outcome & Engineering Decision |
|---|---|---|---|---|
| **Video Face CNN** | **GenD CNN (MIT)**<br>**Shipped Primary** | Generalizable Deepfake detection CNN trained on face boundary artifacts. | **Val AUC @ 360p: 0.877**<br>**FF++ Test AUC: 0.946**<br>**DFDC Test AUC: 0.778** | **SHIPPED as Primary: Won pre-registered tournament by +0.183 validation AUC over SBI baseline under 360p call compression.** |
| **Video Face CNN** | **Our SBI Blend CNN**<br>Our Trained Baseline | Trained from scratch by us (research/video/train_sbi.py) on **real human faces only** via synthetic Self-Blended Images. | FF++ Test AUC: **0.786**<br>DFDC Test AUC: **0.633** | **KEPT as Fallback:** Retained in repository as zero-external-dependency fallback when external model weights are absent. |
| **Video Face CNN** | **Hugging Face Classifiers**<br>(Community Forensics) | Popular off-the-shelf binary 'deepfake detection' image models on Hugging Face. | **FF++ Test AUC: 0.52 – 0.65 (Barely above random chance!)** | **REJECTED:** Proved our core thesis that popular off-the-shelf classifiers overfit to uncompressed lab data and collapse on real call compression. |
| **Voice Clone Scorer** | **Wav2Vec2-XLSR (Gustking)**<br>**Shipped Primary** | Exported to 1.3 GB ONNX trunk with single folded linear head (Apache-2.0). | **Val AUC @ Opus: 0.949**<br>**In-the-Wild Clean: 0.952**<br>**In-the-Wild Opus: 0.947** | **SHIPPED as Primary: Won pre-registered voice tournament by +0.157 val AUC under Opus 16kbps over our re-trained model.** |
| **Voice Clone Scorer** | **Our Re-trained Recipe**<br>Our Trained Model | Published deepfake recipe re-trained by us on strictly licensed speech datasets (research/voice/bench.py). | In-domain CV AUC: **1.000**<br>Val AUC under Opus: **0.791** | **HONESTLY CONCEDED:** In-domain was perfect (1.000), but generalization to real-world out-of-domain Opus fell to 0.791. Protocol dictated shipping Gustking. |
| **Text Scam Classifier** | **TF-IDF + De-obfuscator**<br>**Shipped Primary** | Our custom de-obfuscation pipeline + n-gram Logistic Regression (config/text_model.npz). | **Catch Rate: 96.6%**<br>**Obfuscated Catch: 96.2%**<br>**Size: 6 MB · Latency: 12 ms** | **SHIPPED: Extremely fast, lightweight, and de-obfuscator preserves recall under adversarial leet/homoglyph attacks.** |
| **Text Scam Classifier** | **Multilingual-E5-Small**<br>Transformer Baseline | State-of-the-art multilingual sentence transformer embeddings + LR classifier. | Catch Rate: **96.8%**<br>Size: **150+ MB**<br>Latency: **85 ms** | **REJECTED for Production:** +0.2% recall gain did not justify a 7× latency penalty and 25× model memory footprint for on-device consumer laptops. |

## 5. Empirical Scoreboard: The Full Evidence Ledger

Every metric reported below is derived from pre-registered held-out evaluation splits. No training data was leaked, and thresholds were frozen on validation sets prior to testing:

| Dataset & Split | Sample Size (n) | Evaluated Condition | Catch / Recall Rate | False Positive Rate | 95% Confidence Interval |
|---|---|---|---|---|---|
| **7 Open Text Datasets**<br>(UCI SMS, Phishing, etc.) | 6,576 texts<br>(2,213 scams, 4,363 legit) | Clean text, held-out sources | **96.6% (2,137 / 2,213)** | **2.7% (0.9% at HOLD)** | [95.8%, 97.3%] catch<br>[2.2%, 3.2%] false alarm |
| **Attacked Text Benchmark**<br>(Homoglyphs, leet, spaces) | 2,213 attacked scam texts | Combined obfuscation attack | **96.2% (with de-obf)**<br>**vs 40.1% (no layer)** | Unchanged | [95.4%, 97.0%] with layer<br>[38.1%, 42.2%] no layer |
| **FaceForensics++ c23**<br>(Official identity split) | 810 test videos<br>(695 fake, 115 real) | Original resolution (orig) | **74.7% flagged for call-back** | **0 / 115 real called fake (0%)** | AUC: **0.946** [0.932, 0.960] |
| **FaceForensics++ c23**<br>(Call simulation 720p) | 810 test videos | VP8 1.2Mbps + JPEG q70 | **58.8%** flagged for call-back | **2.2%** on genuine videos | AUC: **0.913** [0.895, 0.931] |

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|---|---|---|---|---|---|
| **FaceForensics++ c23** (Call simulation 360p) | 810 test videos | VP8 600kbps + JPEG q50 | **50.6%** flagged for call-back | **5.0%** on genuine videos | AUC: **0.877** [0.854, 0.900] |
| **DFDC Part 0–2** (Unseen dataset) | 623 test videos (250 fake, 373 real) | Wild unseen lighting/faces | **24.8%** flagged for call-back | **0.7% on real videos** | AUC: **0.778** [0.734, 0.823] |
| **In-the-Wild Voice** (Real-world celebrity fakes) | 315 clips with >= 3s speech (155 fake, 160 real) | Clean & Opus 16 kbps VoIP | **63.9% (Clean)** **66.5% (Opus)** | **0.6% on real voices** | Clean AUC: **0.952** [0.91, 0.98] Opus AUC: **0.947** [0.90, 0.97] |
| **Mixed Spot-Check Kit** (Full Product on 184 files) | 184 files (80 text, 40 photo, 40 voice, 24 video) | Whole product pipeline via satya scan | **68 / 96 fakes caught (70.8%)** | **86 / 88 genuine passed (97.7%)** | Total runtime: ~7.5 min on 4-core CPU (sha256 verified) |
| **SME Held-out Payments** (Accounts Payable cases) | 298 total payment cases (98 attacks, 200 routine) | Deterministic floor + raise policy | **78.6% attacks >= VERIFY** **100% payee changes** | **3.5% routine at HOLD** | Attacks at HOLD: 16.3% Payee change catch: 17/17 (100%) |

## 6. Adversarial Attack Probes & Documented Blind Spots

**Scientific credibility requires testing your own failure modes.** We subjected Satyapan to deliberate adversarial evasion attacks. Here is what worked, what failed, and how our architecture compensates:

| Attacker Move | Observed Experimental Impact | Why it Succeeded / Failed | Engineering Mitigation in Shipped System |
|---|---|---|---|
| **1. Background Noise Injection** (Voice Clone) | **Adding 10 dB SNR white noise collapses voice clone catch from 63.9% → 1.9% (AUC drops 0.952 → 0.843).** | Wav2Vec2 self-attention layers attend to acoustic background artifacts; noise masks synthetic neural vocoder spectral discontinuities. | **Mitigation:** Tested an afftdn denoiser in front; it worsened AUC (-0.010), so rejected. Instead, we **fail-closed**: noisy calls trigger VERIFY, and A1-A7 speech asks + payment floor carry the decision. |
| **2. WhatsApp Re-encoding** (Photo Check) | Forwarding fake face photos via WhatsApp (JPEG q60) drops AUC from **0.913** → **0.872**; catch drops 49% → 35%. | High-frequency 8x8 DCT quantization in JPEG compression discards subtle pixel interpolation seams. | **Mitigation:** Real photos maintain 0.0% false alarms. Companion UI explicitly detects compressed transfers and asks user for the uncompressed file. |
| **3. Pitch & Tempo Shifts** (Voice Clone) | **Altering clone audio (+10% speed, +2 semitones pitch) has zero evasion benefit (AUC delta: -0.004).** | Wav2Vec2 self-supervised pre-training across 53 languages learns invariant phonetic representations unaffected by pitch/tempo shifts. | **No change needed:** Model is inherently robust against pitch/speed manipulation. |
| **4. Text Obfuscation** (Homoglyphs/Leet) | Replacing letters with Cyrillic lookalikes or zero-width joiners collapses vanilla TF-IDF recall from **96.6%** → **40.1%**. | Subword tokenizers fragment modified words into unknown out-of-vocabulary tokens. | **Mitigation:** Our custom pre-filtering NFKD normalizer, zero-width stripper, and leet mapper restores recall to **96.2%**. |
| **5. Unseen Synthesis Models** (FaceShifter) | **Unseen generator (FaceShifter) caught at only 31.7% vs 85.4% on seen manipulation methods.** | FaceShifter uses occlusion-aware facial synthesis that minimizes boundary warping artifacts. | **Mitigation:** Multi-modal fusion. Even when the face model misses, speech asks and invoice diffs escalate to VERIFY. |
| **6. Hinglish Language Bias** (Genuine Texts) | **Genuine Hinglish texts experience a 6.2% false alarm rate (compared to 0.0% in Hindi and 3.2% in English).** | Public Indian fraud training sets over-represent Hinglish scam call scripts, biasing the n-gram priors. | **Mitigation:** Language signals are capped at VERIFY; user is given a 'Call back on known number' button rather than a freeze. |

## 7. IIT Delhi Judging Rubric: Direct Answers & Scoring Proof

| Criterion & Weight | Satyapan's Defensible Answer | Verification Proof & Test Command |
|---|---|---|
| **1. Working Prototype** (25 pts) | End-to-end multi-modal pipeline runs in real-time. Text risk in 12ms; screen OCR in seconds; live audio transcription ~1s behind speech on device. All 4 companion windows interactively functional. | `make live (engine)`<br>`cd desktop && npm run dev`<br>`docs/JUDGES.md §1` |
| **2. Measured Accuracy** (30 pts) | Evaluated across 7 text sets, FF++, DFDC, and In-the-Wild voice. 96.6% scam text caught, 0/563 real videos false-alarmed. 184-file spot kit achieves 97.7% pass on genuine and 70.8% catch on fakes. | `make judge (0.3s scoreboard)`<br>`results/EVIDENCE_LEDGER.md`<br>`docs/BENCHMARKS.md` |
| **3. Own Engineering** (15 pts) | Policy engine with raise-only invariants, A1-A7 ask extractor, de-obfuscation pipeline, 2-stage face tracker, SBI blend CNN fallback, OOB LAN mobile protocol, Section 63 BSA legal report generator. | `docs/ENGINEERING_DEEP_DIVE.md`<br>`src/satyapan/policy/`<br>`docs/JUDGES.md §3` |
| **4. Alert & Safeguards** (15 pts) | Zero mic access (system loopback only); memory-only buffers; coarse bands hide probabilities; no automated reports submitted (MANUAL_ONLY drafts); network strictly quarantined. | `docs/finale/SAFEGUARDS.md`<br>`tests/invariants/ (43 tests)`<br>`docs/JUDGES.md §4` |
| **5. Delivery vs Plan** (15 pts) | Delivered all core Round 1 promises (floor, policy, OOB, report pack, desktop companion). Added: video face check, photo check, OCR, consumer /check page. Cut: AWS cloud deploy (kept on-device for privacy). | `docs/finale/DELIVERED.md`<br>`docs/STATUS.md`<br>`docs/JUDGES.md §5` |
| **6. Problem Understanding** (20 pts) | Addresses ■22,845 Cr cyber fraud epidemic in India (Lok Sabha 2025). Tackles digital arrest, fake CBI, and KYC expiry by anchoring on the un-spoofable ask and the UPI payment floor rather than fighting media quality arms races. | `docs/JUDGES.md §6`<br>`docs/THREAT_MODEL.md` |
| **7. Deployability & Cost** (15 pts) | Three clear deployment paths: (A) Free consumer desktop app / web check, (B) Bank/UPI SDK integration (< ■1/check compute cost on bank cloud), (C) Cyber Cell 1930 triage kit running locally on a police laptop. | `docs/CROSS_PLATFORM.md`<br>`docs/JUDGES.md §7` |
| **8. Live User Journey** (25 pts) | Jury-tested live flow: Call incoming → Consent → Real-time Whisper captions → Immediate A1-A7 ask detection → Satya warns 'CBI never demands UPI transfer' → One-click Safe Reply / Call 1930 / Generate Golden-Hour Pack. | `docs/finale/LIVE_DEMO.md`<br>`make live-personal`<br>`scripts/dry_run.py` |
| **9. Responsible AI** (10 pts) | Zero automated blocking; VERIFY routes to an out-of-band call to an existing number. Plain Hindi/English rationales. Disclosed language biases (Hinglish 6.2% error). DPDP-compliant local data sovereignty. | `docs/DATA_POLICY.md`<br>`docs/JUDGES.md §11` |

## 8. Fast-Track Jury Commands & Stage Demo Cue Sheet

| Command | What it Demonstrates to the Judges | Runtime |
|---|---|---|
| `make judge` | Prints the complete measured scoreboard across all pre-registered frozen test sets, fixtures, and failure breakdowns. | **0.3 s** |
| `make dry-run` | Executes the full automated in-process journey: consent → call → live chips → alert → approver OOB confirm → report pack. | **~100 ms** |
| `make oob-check` | Validates the mobile approver web application served over real local HTTP. | **~1.2 s** |
| `make scan F=kit/` | Processes any jury-provided directory of media/text files; outputs an audit CSV with bands, reasons, and missing models. | Variable |
| `make check` | Verifies codebase integrity: Ruff linter + format check + Mypy strict types + all 880 Pytest unit tests. | **~28 s** |

**FINAL DEFENSE STATEMENT:** *"Satyapan does not pretend deepfake detection is solved. Generative AI will continue to improve. Our breakthrough is architectural: by anchoring on the transaction floor and making detectors raise-only, we make sure that no matter how convincing the deepfake is, the money never moves without true verification."*